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STRUCTURE SOUNDNESS EVALUATION USING ELASTIC WAVES ON DIVIDED AREAS

Abstract

According to one embodiment, a structure evaluation system of the embodiments includes a plurality of sensors, a position locator, and an evaluator. The plurality of sensors detect elastic waves. The position locator locates elastic wave sources on the basis of a plurality of elastic waves detected by each of the plurality of sensors. The evaluator evaluates a deterioration state of a structure in accordance with a presence or absence of a sensor and an elastic wave source distribution which shows a position of the elastic wave sources obtained on the basis of a result of locating of the elastic wave sources.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of and claims benefit under 35 U.S.C. § 120 to U.S. application Ser. No. 17/184,667, filed on Feb. 25, 2021, which is a Continuation Application of International Application PCT/JP2018/034004, filed on Sep. 13, 2018; the entire contents of all of which are incorporated herein by reference.

FIELD

[0002] Embodiments of the present invention relate to a structure evaluation system, a structure evaluation apparatus, and a structure evaluation method.

BACKGROUND

[0003] It is possible to detect elastic waves generated from a damaged portion inside a structure by installing a sensor (for example, an acoustic emission (AE) sensor) on a surface of the structure such as a bridge or bedrock. By installing a plurality of sensors on the surface of a structure, a source location of generated elastic wave (hereinafter, referred to as “elastic wave source”) can be located based on a difference in arrival times of the elastic waves at the sensors. Moreover, since elastic waves are also generated by applying impacts to the structure from the outside, the position of an elastic wave source can be located. However, if there is damage in a propagation path of elastic waves, transmission of elastic waves is hindered, and thus the accuracy of detection of elastic waves by a sensor is reduced. As a result, a locating accuracy of an elastic wave source also decreases. It is possible to detect damage inside a structure from a disorder of a distribution of elastic wave sources whose positions are located by using features of such a structure.

[0004] However, the position of an elastic wave source may be erroneously evaluated depending on a relationship between an area with the damage inside the structure and the installation position of a sensor. As a result, an evaluation accuracy of a deterioration state of a structure may decrease in some cases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram which shows a configuration of a structure evaluation system according to a first embodiment.

[0006] FIG. 2 is a diagram which shows an elastic wave source distribution generated by a position locator locating a position of an elastic wave source when uniform impacts are applied to the structure.

[0007] FIG. 3 is a contour diagram of the elastic wave source distribution in FIG. 2.

[0008] FIG. 4A is a cross-sectional view of a structure when a sensor is installed immediately under damage.

[0009] FIG. 4B is a cross-sectional view of a structure when a sensor is not installed immediately under damage.

[0010] FIG. 5 is a flowchart which shows a flow of deterioration state evaluation processing performed by a structure evaluation apparatus according to the first embodiment.

[0011] FIG. 6 is a diagram which shows a configuration of a structure evaluation system according

to a second embodiment.

[0012] FIG. 7 is a flowchart which shows a flow of deterioration state evaluation processing performed by the structure evaluation apparatus according to the second embodiment.

[0013] FIG. 8 is a diagram which shows a configuration of a structure evaluation system according to a third embodiment.

[0014] FIG. 9 is a diagram for describing processing of a corrector included in a structure evaluation apparatus according to the third embodiment.

[0015] FIG. 10 is a flowchart which shows a flow of deterioration state evaluation processing performed by the structure evaluation apparatus according to the third embodiment.

DETAILED DESCRIPTION

[0016] The present invention provides a structure evaluation system, a structure evaluation apparatus, and a structure evaluation method which can improve an evaluation accuracy of a deterioration state of a structure.

[0017] According to one embodiment, a structure evaluation system of embodiments includes a plurality of sensors, a position locator, and an evaluator. The plurality of sensors detect elastic waves. The position locator locates elastic wave sources on the basis of a plurality of elastic waves detected by each of the plurality of sensors. The evaluator evaluates a deterioration state of a structure in accordance with a presence or absence of a sensor and an elastic wave source distribution which shows the position of the generation source obtained on the basis of results of the locating of a generation source of the elastic waves.

[0018] Hereinafter, a structure evaluation system, a structure evaluation apparatus, and a structure evaluation method according to embodiments will be described with reference to the drawings.

First Embodiment

[0019] FIG. 1 is a diagram which shows a configuration of a structure evaluation system **100** according to a first embodiment.

[0020] The structure evaluation system **100** is used for evaluation of soundness of a structure. In the following description, the evaluation means determining a degree of the soundness of a structure, that is, a deterioration state of the structure, on the basis of a certain reference. Note that bridges made of concrete will be described as an example of the structure in the following description, but the structure is not necessarily limited to bridges. For example, the structure may be any structure that generates elastic waves due to generation or development of a crack, or external impacts (for example, rain, artificial rain, and the like). Note that bridges are not limited to structures built on rivers and valleys and also include various structures provided above the ground (for example, viaducts on expressways).

[0021] In addition, damage that affects an evaluation of the deterioration state of a structure includes, for example, damage inside a structure which interferes with transmission of elastic waves such as cracks, void, disintegration, and, scaling. Here, cracks include vertical cracks, horizontal cracks, diagonal cracks, and the like. Vertical cracks are cracks that are generated in a direction perpendicular to the surface of a structure on which sensors are installed. Horizontal cracks are cracks that are generated in a horizontal direction with respect to the surface of a structure on which sensors are installed. Diagonal cracks are cracks that are generated in directions other than the horizontal and vertical directions with respect to the surface of a structure on which sensors are installed. Disintegration and scaling having a state of mixture of sands and gravels is caused by deterioration at boundary between asphalt and a concrete slab.

[0022] The structure evaluation system **100** includes an impact imparting unit **10**, a plurality of sensors **20-1** to **20-n** (n is an integer equal to or more than 2), a signal processor **30**, and a structure evaluation apparatus **40**. The signal processor **30** and the structure evaluation apparatus **40** are connected to be able to communicate in a wired manner or wirelessly. In the following description, if there is no distinction between the sensors **20-1** to **20-n**, the sensors will be referred to as a sensor **20**.

[0023] The impact imparting unit **10** generates elastic waves inside the structure by giving an impacts **11** to a structure **50**. The impact imparting unit **10** is, for example, a device provided in a vehicle traveling on the structure **50**. The impact imparting unit **10** applies a large number of impacts **11** to a road surface of the structure **50** in a uniform distribution. The impacts **11** may be applied to a wide range of the road surface of the structure **50**. For example, the impacts **11** are applied by spraying water droplets, ice particles, or solids, hammering with a hammer, heating with laser, and the like. When the impact imparting unit **10** sprays water droplets as the impacts **11**, it is desirable that a size and timing of a water droplet hitting the road surface can be controlled by adjusting a nozzle and controlling a discharge timing. It is desirable that a strength and timing of the impacts **11** can be controlled such that they have desired values even in consecutive hits with a hammer.

[0024] The sensor **20** is installed in the structure **50**. For example, the sensor **20** is installed on a surface opposite to the surface onto which the impacts **11** is applied by the impact imparting unit **10**. The sensor **20** has a piezoelectric element, detects elastic waves generated from the inside the structure **50**, and converts the detected elastic waves into an AE source signal that is a voltage signal. The sensor **20** performs processing such as amplification and frequency limitation on the AE source signal, and outputs the processed signal to the signal processor **30**. Note that an acceleration sensor may be used instead of the sensor **20**. In this case, the acceleration sensor performs the same processing as that of the sensor **20** and outputs the processed signal to the signal processor **30**. A thickness of the structure **50** is, for example, 15 cm or more.

[0025] The signal processor **30** inputs the AE source signal processed by the sensor **20**. The signal processor **30** performs signal processing such as required noise removal and parameter extraction on the input AE source signal, and extracts an AE feature amount including information on elastic waves from the input AE source signal. The information on elastic waves is, for example, information such as an amplitude, energy, a rise time, a duration, a frequency, and the number of zero cross counts. The signal processor **30** outputs information based on the extracted AE feature amount to the structure evaluation apparatus **40** as an AE signal. The AE signal output by the signal processor **30** includes information such as a sensor ID, an AE detection time, an AE source signal amplitude, an energy, a rise time, and a frequency. The AE detection time represents a time at which elastic waves are detected by the sensor **20**.

[0026] The signal processor **30** performs, for example, signal processing such as noise removal and parameter extraction. The information on elastic waves is, for example, information on the amplitude, the energy, the rise time, the duration, the frequency, and the number of zero cross counts. The AE signal output by the signal processor **30** includes information such as a sensor ID, an AE detection time, an AE source signal amplitude, an energy, a rise time, and a frequency.

[0027] Here, the amplitude of an AE source signal is, for example, a value of the maximum amplitude among elastic waves. The energy is, for example, a value obtained by time-integrating squares of the amplitude at each time point. Note that a definition of energy is not limited to the example described above, and may be approximated using, for example, an envelope waveform. The rise time is, for example, a time required for elastic waves to rise from a zero value to a value exceeding a predetermined value set in advance. The duration is, for example, a time from a start of a rise of elastic waves until the amplitude becomes smaller than a value set in advance. The frequency is a frequency of elastic waves. The number of zero cross counts is, for example, the number of times elastic waves cross a reference line passing through the zero value.

[0028] The structure evaluation apparatus **40** includes a central processing unit (CPU), a memory, an auxiliary storage device, and the like connected by a bus, and executes an evaluation program. The structure evaluation apparatus **40** functions as a device including a position locator **401**, an evaluator **402**, and a display **403** by executing the evaluation program. Note that all or some of functions of the structure evaluation apparatus **40** may be realized using hardware such as an application specific integrated circuit (ASIC), a programmable logic device (PLD), and a field

programmable gate array (FPGA). In addition, the evaluation program may be recorded on a computer-readable recording medium. The computer-readable recording medium is, for example, a flexible disk, a magneto-optical disc, a ROM, a portable medium such as a CD-ROM, or a storage device such as a hard disk embedded in a computer system. In addition, the evaluation program may be transmitted and received via a telecommunication line.

[0029] The position locator **401** inputs the AE signal output from the signal processor **30**. In addition, the position locator **401** holds information on an installation position of the sensor **20** installed in the structure **50** (hereinafter, referred to as “sensor position information”) in association with the sensor ID in advance. The information on an installation position is, for example, a latitude and longitude, or a distance in a horizontal direction and a vertical direction from a reference position of the structure **50**.

[0030] The position locator **401** locates a position of an elastic wave sources on the basis of information such as a sensor ID and an AE detection time included in the input AE signal, and the sensor position information held in advance. Specifically, the position locator **401** locates the position of elastic wave sources on the basis of a difference of an arrival time of elastic waves at the plurality of sensors **20**, that is, the AE detection time. The position locator **401** generates an elastic wave source distribution in an evaluation target area in the structure **50** using information on the positions of a plurality of elastic wave sources obtained during a certain period of time. The evaluation target area in the structure **50** is an area surrounded by the sensor **20**. The elastic wave source distribution is a distribution indicating the located positions of the elastic wave sources. The position locator **401** outputs the generated elastic wave source distribution to the evaluator **402**.

[0031] The evaluator **402** inputs the elastic wave source distribution output from the position locator **401**. The evaluator **402** evaluates a deterioration state of the structure **50** on the basis of the elastic wave source distribution and the installation position of the sensor **20**. Specifically, the evaluator **402** acquires a feature amount of elastic waves in the elastic wave source distribution on the basis of the elastic wave source distribution, and evaluates the deterioration state of the structure **50** using the acquired feature amount and the installation position of the sensor **20**. The feature amount of elastic waves in the elastic wave source distribution is, for example, a density of an elastic wave source, a volume of elastic waves, and an amplitude of elastic waves.

[0032] As specific processing, first, the evaluator **402** divides a range in which the elastic wave source distribution extends into areas suitable for evaluation (hereinafter, referred to as “evaluation areas”). The evaluation areas are set in advance. For example, the evaluation areas are set to be equal to or smaller than a size of damage to be detected and to include one or more positions to which the impacts **11** is applied. It is possible to evaluate a presence or absence of a smaller damage when a space between the areas is narrower. In addition, as a position to which more impacts **11** are applied is included in one area, a difference in density is significant, and an evaluation accuracy is further improved.

[0033] Next, the evaluator **402** calculates the density of an elastic wave source for each evaluation area. Then, the evaluator **402** evaluates the deterioration state of the structure **50** by comparing the calculated density of an elastic wave source for each evaluation area with at least one of a first threshold and a second threshold determined in advance. The second threshold is larger than the first threshold. The evaluator **402** causes the display **403** to display a result of the evaluation.

[0034] The display **403** is an image display device such as a liquid crystal display or an organic electro luminescence (EL) display. The display **403** displays the result of the evaluation according to control of the evaluator **402**. The display **403** may be an interface for connecting an image display device to the structure evaluation apparatus **40**. In this case, the display **403** generates a video signal for displaying the result of the evaluation and outputs the video signal to the image display device connected thereto.

[0035] Next, a reason for setting the first threshold and the second threshold as described above will be described.

[0036] FIG. 2 is a diagram which shows the elastic wave source distribution generated by the position locator **401** performing the locating of an elastic wave sources when uniform impacts **11** are applied to the structure **50**. In addition, FIG. 3 is a contour diagram of the elastic wave source distribution in FIG. 2. In FIGS. 2 and 3, the vertical and horizontal axes represent a length (m) of an evaluation target area. Positions of black circles in FIGS. 2 and 3 indicate the installation position of the sensor **20**. FIGS. 2 and 3 show a case in which 15 sensors **20** are installed. A position of a triangle **55** in FIG. 2 shows the position of elastic wave sources.

[0037] It is assumed that there is a damage in areas **60-1** and **60-2** shown in FIG. 2. Although the areas **60-1** and **60-2** have damage, it is evaluated that the number of elastic wave sources is relatively less in the area **60-1** than in the area **60-2** in the example shown in FIG. 2. A difference between these two areas is that the sensor **20** is installed immediately below the damage in the area **60-1** and the sensor **20** is not installed immediately below the damage in the area **60-2**. Note that each grid-shaped area **56** indicated by dotted lines is an evaluation area in FIG. 2.

[0038] With reference to FIG. 3, while the density of the elastic wave source due to the damage is significantly reduced in the area **60-1**, the density of the elastic wave source due to the damage is relatively large in the area **60-2**. Therefore, when the evaluator **402** has evaluated the areas **60-1** and **60-2** using the same one threshold, if the threshold is too low, the density of the elastic wave source of a damaged portion in the area **60-2** does not fall below the threshold and it is not evaluated as damage. On the other hand, if the threshold is too large, when the density of elastic waves has a slight difference due to a variation of an execution method at the time of measurement such as an impact application condition, a measurement condition, or the like in the area **60-1**, the damaged portion is erroneously evaluated as a soundness area.

[0039] A reason for occurrence of the phenomenon described above will be described using FIG. 4.

[0040] FIG. 4A is a cross-sectional view of the structure **50** when the sensor **20** is installed immediately below damage **62**. FIG. 4B is a cross-sectional view of the structure **50** when the sensor **20** is not installed immediately below the damage **62**. In FIGS. 4A and 4B, damage in the horizontal direction is shown as an example of the damage **62**. However, the damage in the present embodiment is a crack, a void, disintegration, and, scaling, or the like as described above. In FIGS. 4A and 4B, a reference numeral **61** indicates an application position of the impacts **11**. Since a transmission of elastic waves generated on an upper surface of the structure **50** due to the impacts **11** applied at an application position **61** is hindered by the damage **62**, the elastic waves hardly transmit the damage **62**. Therefore, the elastic waves diffracted by the damage **62** reach the sensor **20**.

[0041] In a case of FIG. 4A, since a diffraction angle of elastic waves is large, an attenuation is large, and the elastic waves hardly reach the sensor **20** immediately below the damage **62**. In addition, even if the elastic waves reach the sensor **20**, since a difference in arrival times of elastic waves at each of the sensors **20-1** and **20-2** shown in FIG. 4A is equal to a time difference from the end of the damage **62**, an elastic wave source **55** is located at the end of the damage **62**. As described above, when the sensor **20** is installed immediately below the damage **62**, the density of an elastic wave source in an area in which the damage **62** occurs significantly decreases.

[0042] On the other hand, in the case of FIG. 4B, the diffraction angle is relatively small and many elastic waves are diffracted and reach the sensor **20**. In addition, a difference in arrival time of elastic waves at each of the sensors **20-1** and **20-2** shown in FIG. 4B may not be much different from a case in which there is no damage **62**. For this reason, the number of elastic wave sources located at a position immediately above the damage **62** increases. As described above, when the sensor **20** is not installed immediately below the damage **62**, a decrease in the density of an elastic wave source in the area in which a damage occurs is smaller than when the sensor **20** is installed immediately below the damage **62**.

[0043] As described above, a decrease degree of the density of an elastic wave source varies depending on whether or not the sensor **20** is installed below the area in which the damage **62**

occurs. Therefore, the structure evaluation apparatus **40** in the present embodiment evaluates the deterioration state of the structure **50** on the basis of the elastic wave source distribution and the installation position of the sensor **20**.

[0044] FIG. **5** is a flowchart which shows a flow of deterioration state evaluation processing performed by the structure evaluation apparatus **40** according to the first embodiment.

[0045] The position locator **401** acquires an AE signal output from the signal processor **30** (step **S101**). The position locator **401** locates elastic wave sources on the basis of information such as a sensor ID, an AE detection time and the like included in the acquired AE signal, and sensor position information held in advance (step **S102**). Thereafter, the position locator **401** generates an elastic wave source distribution using a result of locating the elastic wave sources (step **S103**).

[0046] The position locator **401** outputs the generated elastic wave source distribution and the information on the installation position of the sensor **20** to the evaluator **402**. The evaluator **402** acquires the elastic wave source distribution and the information on the installation position of the sensor **20** output from the position locator **401**. The evaluator **402** sets an evaluation area for the acquired elastic wave source distribution (step **S104**). For example, the evaluator **402** sets an evaluation area (an area **56** shown in FIG. **2**) for the elastic wave source distribution as shown in FIG. **2**.

[0047] The evaluator **402** selects one evaluation area among a plurality of set evaluation areas (step **S105**). At this time, the evaluator **402** selects an evaluation area on which comparison with the threshold is not performed. Next, the evaluator **402** derives the density of an elastic wave source in the selected evaluation area (step **S106**). The evaluator **402** determines whether or not the derived density of an elastic wave source (a first feature amount) is equal to or larger than the first threshold (step **S107**). When the density of an elastic wave source is equal to or larger than the first threshold (YES in step **S107**), the evaluator **402** determines whether or not the derived density of an elastic wave source (the first feature amount) is equal to or larger than the second threshold (step **S108**).

[0048] When the density of an elastic wave source is equal to or larger than the second threshold (YES in step **S108**), the evaluator **402** evaluates the selected evaluation area as a soundness area (step **S109**). The evaluator **402** determines whether or not all evaluation areas have been evaluated (step **S110**). When all the evaluation areas have been evaluated (YES in step **S110**), the evaluator **402** outputs results of the evaluation to the display **403**. The display **403** displays the result of evaluation under control of the evaluator **402** (step **S111**). Specifically, the display **403** superimposes and displays the result of evaluation (soundness or deteriorated) for each evaluation area of the elastic wave source distribution.

[0049] In the processing of step **S110**, when all the evaluation areas have not been evaluated (NO in step **S110**), the evaluator **402** executes processing of step **S105** and thereafter.

[0050] In addition, in the processing of step **S107**, when the density of an elastic wave source is less than the first threshold (NO in step **S107**), the evaluator **402** evaluates the selected evaluation area as a deteriorated area (step **S112**).

[0051] In addition, in the processing of step **S108**, when the density of an elastic wave source is less than the second threshold (NO in step **S108**), the evaluator **402** determines whether or not the sensor **20** is installed in the selected evaluation area using the information on the installation position of the sensor **20** (step **S113**).

[0052] When the sensor **20** is installed in the selected evaluation area (YES in step **S113**), the evaluator **402** evaluates the selected evaluation area as a soundness area (step **S109**).

[0053] On the other hand, when the sensor **20** is not installed in the selected evaluation area (NO in step **S113**), the evaluator **402** evaluates the selected evaluation area as a deteriorated area (step **S112**).

[0054] According to the structure evaluation system **100** configured in this manner, the structure evaluation apparatus **40** evaluates the deterioration state in accordance with the density of an elastic

wave source in an evaluation area and a presence or absence of the sensor **20** in the evaluation area, and thereby it is possible to improve the evaluation accuracy of the deterioration state of a structure. Specifically, the structure evaluation apparatus **40** determines whether or not the sensor **20** is installed in the evaluation area when the density of an elastic wave source in the evaluation area is less than the second threshold. The density of an elastic wave source tends to decrease near a place in which the sensor **20** is installed. Therefore, the structure evaluation apparatus **40** evaluates the evaluation area as a soundness area when the sensor **20** is installed in the evaluation area even if the density of an elastic wave source is less than the second threshold and conditions for evaluating an area as a deteriorated area are satisfied. Then, when the density of an elastic wave source is less than the second threshold and the sensor **20** is not installed in the evaluation area, the structure evaluation apparatus **40** evaluates the evaluation area as a deteriorated area. With such a configuration, it is possible to curb erroneous evaluation due to the installation position of the sensor **20**. For this reason, the evaluation accuracy of the deterioration state of a structure can be improved.

[0055] Moreover, the structure evaluation apparatus **40** evaluates the deterioration state using two thresholds such as a first threshold and a second threshold. For example, when a damage is in the entire structure **50**, the structure evaluation apparatus **40** determines an entire evaluation area as a deteriorated area by comparing the first threshold with the density in the evaluation area. As a result, the structure evaluation apparatus **40** can determine that the evaluation area is a deteriorated area before comparing the second threshold with the density of the evaluation area. For this reason, the evaluation accuracy can be further improved compared to a case of evaluating the deterioration state of a structure using only the second threshold.

[0056] Hereinafter, a modification of the structure evaluation system **100** according to the first embodiment will be described.

[0057] A configuration in which the structure evaluation apparatus **40** compares the first threshold and the second threshold with the density of an elastic wave source which is one of features amounts of elastic waves in the elastic wave source distribution is shown in the present embodiment, but processing of the structure evaluation apparatus **40** is not necessarily limited thereto. For example, the structure evaluation apparatus **40** may be configured to compare the amplitude of elastic waves which is one of feature amounts of elastic waves in the elastic wave source distribution (the second feature amount) with the first threshold for the amplitude of elastic waves, and compare the second threshold with the density of an elastic wave source (the first feature amount). When the amplitude of elastic waves is used, the evaluator **402** derives an average value of the amplitudes of elastic waves for each evaluation area, and compares the derived average value of the amplitudes with the first threshold.

[0058] The structure evaluation apparatus **40** may be configured to evaluate the deterioration state of the structure **50** on the basis of one threshold and the installation position of the sensor **20**. With such a configuration, the structure evaluation apparatus **40** executes the processing of step **S108** after the processing of step **S106** without executing processing of step **S107**.

Second Embodiment

[0059] In a second embodiment, a structure evaluation system evaluates the deterioration state of a structure in accordance with whether or not an evaluation area is included in a predetermined range centering on each sensor.

[0060] FIG. **6** is a diagram which shows a configuration of a structure evaluation system **100a** according to the second embodiment.

[0061] The structure evaluation system **100a** includes an impact imparting unit **10**, a plurality of sensors **20-1** to **20-n**, a signal processor **30**, and a structure evaluation apparatus **40a**. The structure evaluation system **100a** is different from the structure evaluation system **100** in configuration in that it includes a structure evaluation apparatus **40a** instead of the structure evaluation apparatus **40**. In the following description, only the structure evaluation apparatus **40a** will be described.

[0062] The structure evaluation apparatus **40a** includes a CPU, a memory, an auxiliary storage device, and the like which are connected by a bus, and executes an evaluation program. The structure evaluation apparatus **40a** functions as a device including the position locator **401**, an evaluator **402a**, and the display **403** by executing the evaluation program. Note that all or some of functions of the structure evaluation apparatus **40a** may be realized using hardware such as an ASIC, a PLD, or FPGA. In addition, the evaluation program may be recorded on a computer-readable recording medium. The computer-readable recording medium is, for example, a flexible disk, a magneto-optical disc, a ROM, a portable medium such as a CD-ROM, and a storage device such as a hard disk embedded in a computer system. The evaluation program may be transmitted or received via a telecommunication line.

[0063] The structure evaluation apparatus **40a** is different from the structure evaluation apparatus **40** in configuration in that it includes the evaluator **402a** instead of the evaluator **402**. The other constituents of the structure evaluation apparatus **40a** are the same as those of the structure evaluation apparatus **40**. For this reason, only differences between the evaluator **402a** and the evaluator **402** will be described.

[0064] The evaluator **402a** evaluates the deterioration state of the structure **50** in accordance with whether or not the evaluation area is included in a predetermined range centering about each sensor **20** when the density of an elastic wave source is less than the second threshold. The predetermined range centering about the sensor **20** is a range determined with the sensor **20** as a center, and may be, for example, a range with a radius D centering about the sensor **20** or may be a rectangular range having different distances in an X direction and a Y direction, which centers about the sensor **20**. The distance D is set to be sufficiently small with respect to a sensor interval. The evaluator **402a** outputs a result of the evaluation to the display **403**.

[0065] FIG. 7 is a flowchart which shows a flow of the deterioration state evaluation processing performed by the structure evaluation apparatus **40a** according to the second embodiment. Note the same processing as in FIG. 5 will be denoted by the same reference numerals in FIG. 7, and description thereof will be omitted.

[0066] In the processing of step **S108**, when the density of an elastic wave source is less than the second threshold (NO in step **S108**), the evaluator **402a** determines whether or not an evaluation area is included in the predetermined range centering about each sensor **20** (step **S201**). The evaluator **402a** determines that the evaluation area is included in the predetermined range centering about each sensor **20** when some or all of the evaluation area is included in the predetermined range centering about each sensor **20**. The evaluator **402a** determines that the evaluation area is not included in the predetermined range centering about each sensor **20** when all of the evaluation area is not included in the predetermined range centering about the evaluation area.

[0067] When the evaluation area is included in the predetermined range centering about each sensor **20** (YES in step **S201**), the evaluator **402a** evaluates the selected evaluation area as a soundness area (step **S109**).

[0068] On the other hand, when the evaluation area is not included in the predetermined range centering about each sensor **20** (NO in step **S201**), the evaluator **402a** evaluates the selected evaluation area as a deteriorated area (step **S112**).

[0069] The structure evaluation system **100a** configured as described above determines whether or not the evaluation area is included in the predetermined range centering about the sensor **20** when the density of an elastic wave source in the evaluation area is less than the second threshold. As described above, the density of an elastic wave source tends to decrease near the place in which the sensor **20** is installed. Here, even if the density of an elastic wave source is less than the second threshold and conditions for evaluating the evaluation area as a deteriorated area are satisfied, when the evaluation area is included in the predetermined range centering about the sensor **20**, the structure evaluation apparatus **40a** evaluates the evaluation area as a soundness area. Then, when the density of an elastic wave source is less than the second threshold and the evaluation area is not

included in the predetermined range centering about the sensor **20**, the structure evaluation apparatus **40a** evaluates the evaluation area as a deteriorated area. With such a configuration, it is possible to curb an erroneous evaluation due to the installation position of the sensor **20**. For this reason, the evaluation accuracy of the deterioration state of a structure can be improved.

[0070] The structure evaluation system **100a** according to the second embodiment may be modified in a similar manner to the structure evaluation system **100**.

Third Embodiment

[0071] In a third embodiment, a structure evaluation system evaluates a deterioration state of a structure by correcting an elastic wave source density distribution.

[0072] FIG. **8** is a diagram which shows a configuration of a structure evaluation system **100b** according to the third embodiment.

[0073] The structure evaluation system **100b** includes an impact imparting unit **10**, a plurality of sensors **20-1** to **20-n**, a signal processor **30**, and a structure evaluation apparatus **40b**. The structure evaluation system **100b** is different from the structure evaluation system **100** in configuration in that it includes the structure evaluation apparatus **40b** instead of the structure evaluation apparatus **40**. In the following description, only the structure evaluation apparatus **40b** will be described.

[0074] The structure evaluation apparatus **40b** includes a CPU, a memory, an auxiliary storage device, and the like connected by a bus, and executes an evaluation program. The structure evaluation apparatus **40b** functions as a device including the position locator **401**, the evaluator **402b**, the display **403**, and a corrector **404b** by executing the evaluation program. Note that all or some of functions of the structure evaluation apparatus **40b** may be realized using hardware such as an ASIC, a PLD, and an FPGA. In addition, the evaluation program may also be recorded in a computer-readable recording medium. The computer-readable recording medium is, for example, a flexible disk, a magneto-optical disc, a ROM, a portable medium such as a CD-ROM, and a storage device such as a hard disk embedded in a computer system. The evaluation program may be transmitted or received via a telecommunication line.

[0075] The structure evaluation apparatus **40b** is different from the structure evaluation apparatus **40** in configuration in that it includes an evaluator **402b** instead of the evaluator **402** and newly includes the corrector **404b**. The structure evaluation apparatus **40b** is the same as the structure evaluation apparatus **40** in other configurations. For this reason, only the evaluator **402b** and the corrector **404b** will be described.

[0076] The evaluator **402b** generates an elastic wave source density distribution using the derived density of an elastic wave source for each evaluation area and the elastic wave source distribution. The elastic wave source density distribution is a distribution representing the density of an elastic wave source. For example, the elastic wave source density distribution is represented by a contour diagram as shown in FIG. **3**. The evaluator **402b** evaluates the deterioration state of a structure on the basis of a corrected elastic wave source density distribution corrected by the corrector **404b**. The evaluator **402b** outputs a result of the evaluation to the display **403**.

[0077] The corrector **404b** corrects the elastic wave source density distribution by performing different processing on the elastic wave source density distribution according to a distance from the position of the sensor **20**. Specific processing performed by the corrector **404b** will be described using FIG. **9**.

[0078] FIG. **9** is a diagram for describing processing performed by the corrector **404b**.

[0079] As shown in FIG. **9**, the corrector **404b** does not perform correction on an area **70** within a distance **D** from the installation position of the sensor **20** in the areas of the elastic wave source density distribution. In addition, the corrector **404b** performs the following correction on an area **71** that exceeds the distance **D** from the installation position of the sensor **20** in the areas of the elastic wave source density distribution, that is, which is away from the installation position of the sensor **20** by the distance **D** or more in the areas of the elastic wave source density distribution. For example, the corrector **404b** calculates $p' = M - k(M - p)$ with respect to an elastic wave source

density p on the basis of a value of the elastic wave source density obtained in the elastic wave source density distribution and a correction coefficient k , and derives the corrected elastic wave source density p' . Here, M is an average elastic wave source density in a soundness area. In addition, the correction coefficient k and the distance D do not need to be constants, and a coefficient which continuously changes in accordance with measurement conditions such as the sensor interval may also be used.

[0080] FIG. **10** is a flowchart which shows a flow of deterioration state evaluation processing performed by the structure evaluation apparatus **40b** according to the third embodiment. Note that the same processing as in FIG. **5** will be denoted by the same reference numerals in FIG. **10**, and thus description thereof will be omitted.

[0081] After the processing of step **S104**, the evaluator **402b** derives the density of an elastic wave source for each evaluation area (step **S301**). The evaluator **402b** generates an elastic wave source density distribution using the derived density of an elastic wave source for each evaluation area and the elastic wave source distribution (step **S302**). The evaluator **402b** outputs the generated elastic wave source density distribution to the corrector **404b**.

[0082] The corrector **404b** generates a corrected elastic wave source density distribution by correcting the generated elastic wave source density distribution according to the installation position of the sensor **20** (step **S303**). The corrector **404b** outputs the corrected elastic wave source density distribution to the evaluator **402b**. The evaluator **402b** selects one of evaluation areas in the corrected elastic wave source density distribution output from the corrector **404b** (step **S304**). The evaluation area in the corrected elastic wave source density distribution is the same as the evaluation area set in the elastic wave source distribution in the processing of step **S104** by the evaluator **402b**.

[0083] The evaluator **402b** determines whether or not the density of an elastic wave source in the selected evaluation area is equal to or larger than the first threshold (step **S305**). When the density of an elastic wave source is equal to or larger than the first threshold (YES in step **S305**), the evaluator **402b** evaluates the selected evaluation area as a soundness area (step **S306**). The evaluator **402b** determines whether or not all the evaluation areas have been evaluated (step **S307**). When all the evaluation areas have been evaluated (YES in step **S307**), the evaluator **402b** outputs a result of the evaluation to the display **403**.

[0084] In the processing of step **S307**, when all the evaluation areas have not been evaluated (NO in step **S307**), the evaluator **402b** executes processing of step **S304** and thereafter.

[0085] In addition, when the density of an elastic wave source is less than the first threshold in the processing of step **S305** (NO in step **S305**), the evaluator **402b** evaluates the selected evaluation area as a deteriorated area (step **S308**).

[0086] The structure evaluation system **100b** configured in this manner evaluates the deterioration state of the structure **50** by correcting the elastic wave source density distribution according to the distance from the installation position of the sensor **20**. The density indicated by the elastic wave source density distribution is corrected by correcting the elastic wave source density distribution using a correction coefficient. With such a configuration, it is possible to curb erroneous evaluation due to the installation position of the sensor **20**. For this reason, it is possible to improve the evaluation accuracy of the deterioration state of a structure.

[0087] Hereinafter, a common modification in each embodiment will be described.

[0088] The structure evaluation systems **100**, **100a**, and **100b** may include three or more sensors **20**.

[0089] The structure evaluation systems **100**, **100a**, and **100b** may not include the impacts imparting unit **10**. With such a configuration, the impacts **11** to the structure **50** may be manually applied. The impacts **11** manually applied to the structure **50** includes, for example, a spray of water droplets, ice droplets, or solids, consecutive hits with a hammer, heating with laser, and the like.

[0090] The evaluation area does not need to be limited to a grid shape as shown in FIG. 2. Specifically, the evaluation area may have any shape as long as an area is equal to or smaller than the size of damage to be detected and includes one or a plurality of positions to which the impacts **11** is applied. For example, the evaluation area may be a circle, a rectangle having a plurality of vertices (for example, an n-polygon (n is an integer of 3 or more)), or an area designated by handwriting. In addition, the evaluation area may be an area that is equal to or smaller than the size of damage to be detected.

[0091] Some or all of function units of the structure evaluation apparatus **40** may be included in another housing. For example, the structure evaluation apparatus **40** may include only the evaluator **402**, and the position locator **401** and the display **403** may be provided in another housing. With such a configuration, the evaluator **402** acquires the elastic wave source distribution from another housing and evaluates soundness of a structure using the acquired elastic wave source distribution. Then, the evaluator **402** outputs a result of the evaluation to the display **403** provided in another housing.

[0092] With such a configuration, a manufacturing cost of the structure evaluation apparatus **40** can be reduced by using existing device for deriving the elastic wave source distribution.

[0093] The signal processor **30** may be configured to perform signal processing on an AE source signal whose amplitude value is higher than the first threshold determined to be higher than a noise level among the input AE source signals. Specifically, when a large vibration as compared with the first threshold is first detected, the signal processor **30** determines that a signal for a predetermined time from a time at which the vibration exceeds the first threshold as an elastic wave waveform, and stores an AE source signal whose amplitude value is higher than the first threshold. Then, the signal processor **30** extracts an AE feature amount including information on elastic waves on the basis of data of the elastic wave waveform indicated by the stored AE source signal. Note that the first threshold is set in advance.

[0094] The signal processor **30** may be included in the structure evaluation apparatus **40** (or the structure evaluation apparatus **40a** or the structure evaluation apparatus **40b**). With such a configuration, the signal processor **30** acquires an AE source signal processed by the sensor **20** directly or via a relay device (not shown) from the sensor **20**.

[0095] In FIG. 1, one signal processor **30** is connected to the plurality of sensors **20-1** to **20-n**, but the structure evaluation system **100** (or the structure evaluation system **100a** or the structure evaluation system **100b**) may include a plurality of signal processors **30**, and be configured to include a plurality of sensor units in which each signal processor **30** is connected to each sensor **20**.

[0096] In addition, the evaluator **402** (or the evaluator **402a** or the evaluator **402b**) may operate as an output controller. The output controller controls an output unit and outputs a result of the evaluation. Here, the output unit includes the display **403**, a communicator, and a printer. When the output unit is a communicator, the output controller controls the communicator and transmits the result of the evaluation to another device. In addition, when the output unit is a printer, the output controller controls the printer and prints the result of the evaluation. Note that the structure evaluation apparatus **40** (or the structure evaluation apparatus **40a** or the structure evaluation apparatus **40b**) may execute the operation described above by including some or all of the display **403**, the communicator, and the printer as output units.

[0097] According to at least one embodiment described above, it is possible to improve the evaluation accuracy of the deterioration state of a structure by including a plurality of sensors that detect elastic waves, a position locator that locates an elastic wave sources on the basis of a plurality of elastic waves detected by each of the plurality of sensors, and an evaluator that evaluates the deterioration state of a structure in accordance with an elastic wave source distribution obtained on the basis of the elastic wave sources and a presence or absence of a sensor.

[0098] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel

embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A structure evaluation system comprising: a plurality of sensors detecting elastic waves; and one or more processors configured to: locate elastic wave sources on a basis of the elastic waves detected by each of the plurality of sensors; and evaluate a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, wherein the one or more processors is configured to: set a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluate a respective divided area satisfying a third condition among the plurality of divided areas as a soundness area, the third condition being that the first feature amount is less than the predetermined threshold and is included in a predetermined range centering about a at least one of the plurality of sensors, and evaluate a respective divided area satisfying a fourth condition among the plurality of divided areas as deterioration occurring area, the fourth condition being that the first feature amount is less than the predetermined threshold and is not included in the predetermined range centering about a at least one of the plurality of sensors.
2. A structure evaluation system comprising: a plurality of sensors detecting elastic waves; and one or more processors configured to: locate elastic wave sources on a basis of the elastic waves detected by each of the plurality of sensors; and evaluate a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, wherein the one or more processors is configured to: set a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluate a respective divided area satisfying a first condition among the plurality of divided areas as a soundness area, the first condition being that the first feature amount is less than a predetermined threshold value and that at least one sensor is installed, and evaluate a respective divided area satisfying a second condition among the plurality of divided areas as deterioration occurring area, the second condition being that the first feature amount is less than the predetermined threshold and that at least one of the plurality of sensors is not installed.
3. The structure evaluation system according to claim 1, wherein the one or more processors is configured to: compare a second feature amount of elastic waves in the elastic wave source distribution with a threshold different from the predetermined threshold, evaluate a respective area of the plurality of divided areas as deterioration occurring area in case that in the respective area the second feature amount is less than the different threshold, and perform evaluation based on a result of comparison between the first feature amount of the elastic waves with the predetermined threshold in a case that in the respective area the second feature amount is equal to or larger than the different threshold.
4. A structure evaluation apparatus comprising: one or more processors configured to: locate elastic

wave sources on a basis of elastic waves detected by each of a plurality of sensors that detect the elastic waves; evaluate a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, wherein the one or more processor is configured to: set a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluate a respective divided area satisfying a third condition among the plurality of divided areas as a soundness area, the third condition being that the first feature amount is less than the predetermined threshold and is included in a predetermined range centering about a at least one of the plurality of sensors, and evaluate a respective divided area satisfying a fourth condition among the plurality of divided areas as deterioration occurring area, the fourth condition being that the first feature amount is less than the predetermined threshold and is not included in the predetermined range centering about a at least one of the plurality of sensors.

5. A structure evaluation apparatus comprising: one or more processors configured to: locate elastic wave sources on a basis of elastic waves detected by each of a plurality of sensors that detect the elastic waves; evaluate a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, set a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluate a respective divided area satisfying a first condition among the plurality of divided areas as a soundness area, the first condition being that the first feature amount is less than a predetermined threshold value and that at least one sensor is installed, and evaluate a respective divided area satisfying a second condition among the plurality of divided areas as deterioration occurring area, the second condition being that the first feature amount is less than the predetermined threshold and that at least one of the plurality of sensors are not installed.

6. A structure evaluation method comprising: locating elastic wave sources on a basis of elastic waves detected by each of a plurality of sensors that detect the elastic waves; evaluating a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, setting a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluating a respective divided area satisfying a third condition among the plurality of divided areas as a soundness area, the third condition being that the first feature amount is less than the predetermined threshold and is included in a predetermined range centering about a at least one of the plurality of sensors, and evaluating a respective divided area satisfying a fourth condition among the plurality of divided areas as deterioration occurring area, the fourth condition being that the first feature amount is less than the predetermined threshold and is not included in the predetermined range centering about a at least one of the plurality of sensors.

7. A structure evaluation method comprising: locating elastic wave sources on a basis of elastic waves detected by each of a plurality of sensors that detect the elastic waves; evaluating a deterioration state of a structure based on one or more criteria determined according to an installation position of the plurality of sensors in an elastic wave source distribution which shows a

position of the elastic wave sources obtained on a basis of a result of locating of the elastic wave sources, using both information on the installation position of the plurality of sensors and the elastic wave source distribution, setting a plurality of divided areas by dividing the elastic wave source distribution, compare a first feature amount of elastic waves in the elastic wave source distribution with a predetermined threshold, evaluating a respective divided area satisfying a first condition among the plurality of divided areas as a soundness area, the first condition being that the first feature amount is less than a predetermined threshold value and that at least one sensor is installed, and evaluating a respective divided area satisfying a second condition among the plurality of divided areas as deterioration occurring area, the second condition being that the first feature amount is less than the predetermined threshold and that at least one of the plurality of sensors are not installed.
